EEG Analysis Report

I. Patient Data

Patient Name : None

Patient ID : None

Measurement Date : 2019-05-06 14:00:30.520000+00:00

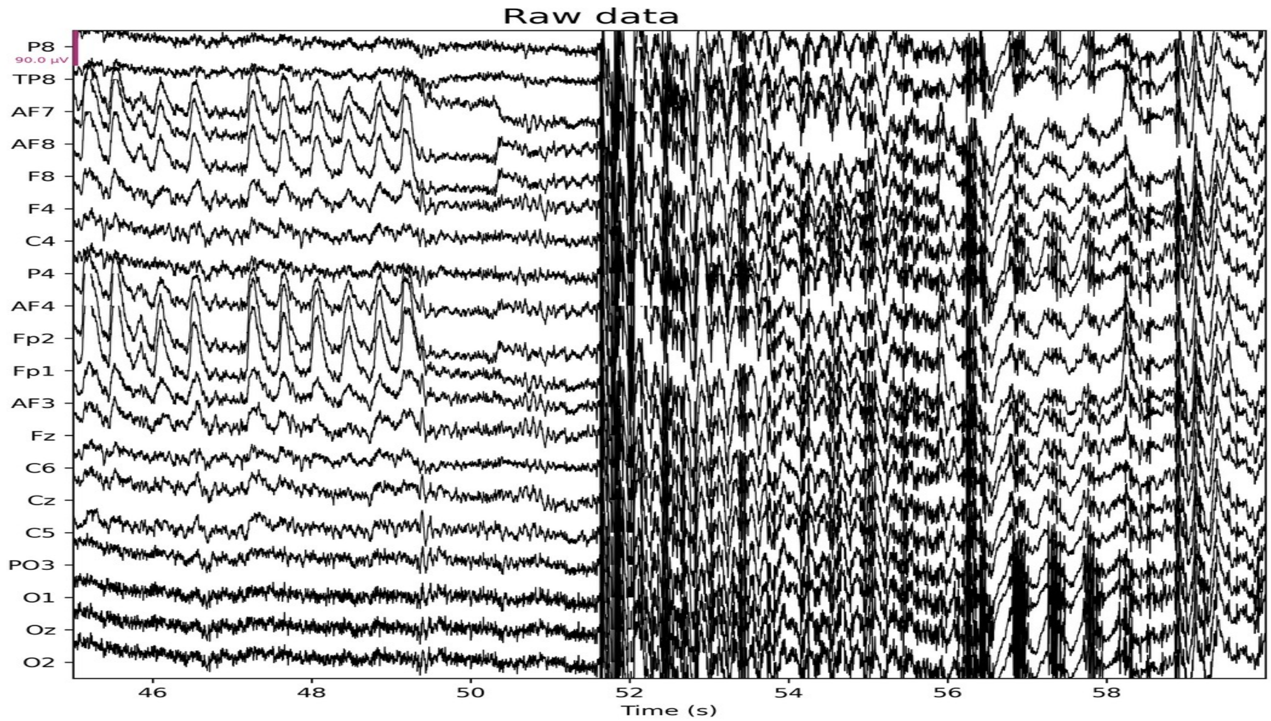
Sampling Frequency : 500.0 Hz

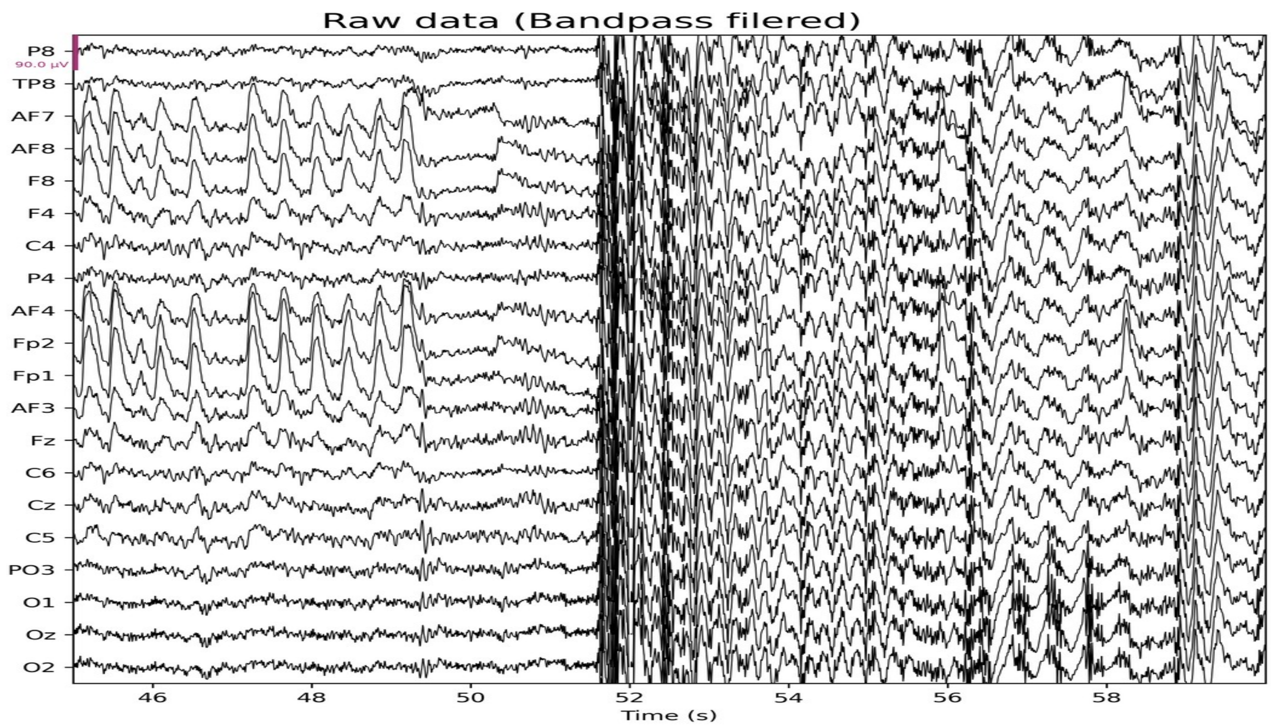
Number of Channels : 33

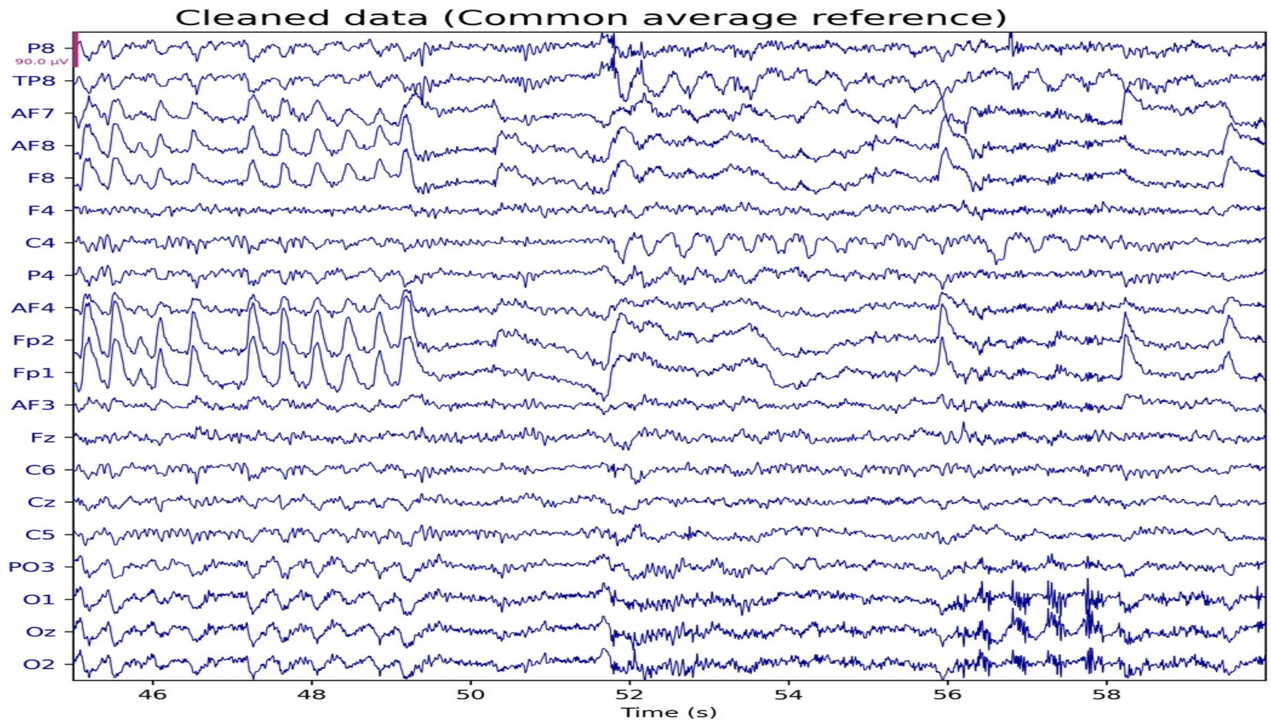
Channel Names : P8, TP8, AF7, AF8, F8, F4, C4, P4, AF4, Fp2, Fp1, AF3, Fz, C6, Cz, C5, PO3, O1, Oz, O2, PO4, Pz, Fpz, FC6, P3, C3, F3, F7, FC5, FCz, TP7, P7, STI 014

II. EEG

EEG data is a record of the oscillations of electrical brain potentials recorded from electrodes on the human scalp (T100) (T101) (T005).The raw data in the figure below have been cleaned by the application of high-pass and low-pass filters.

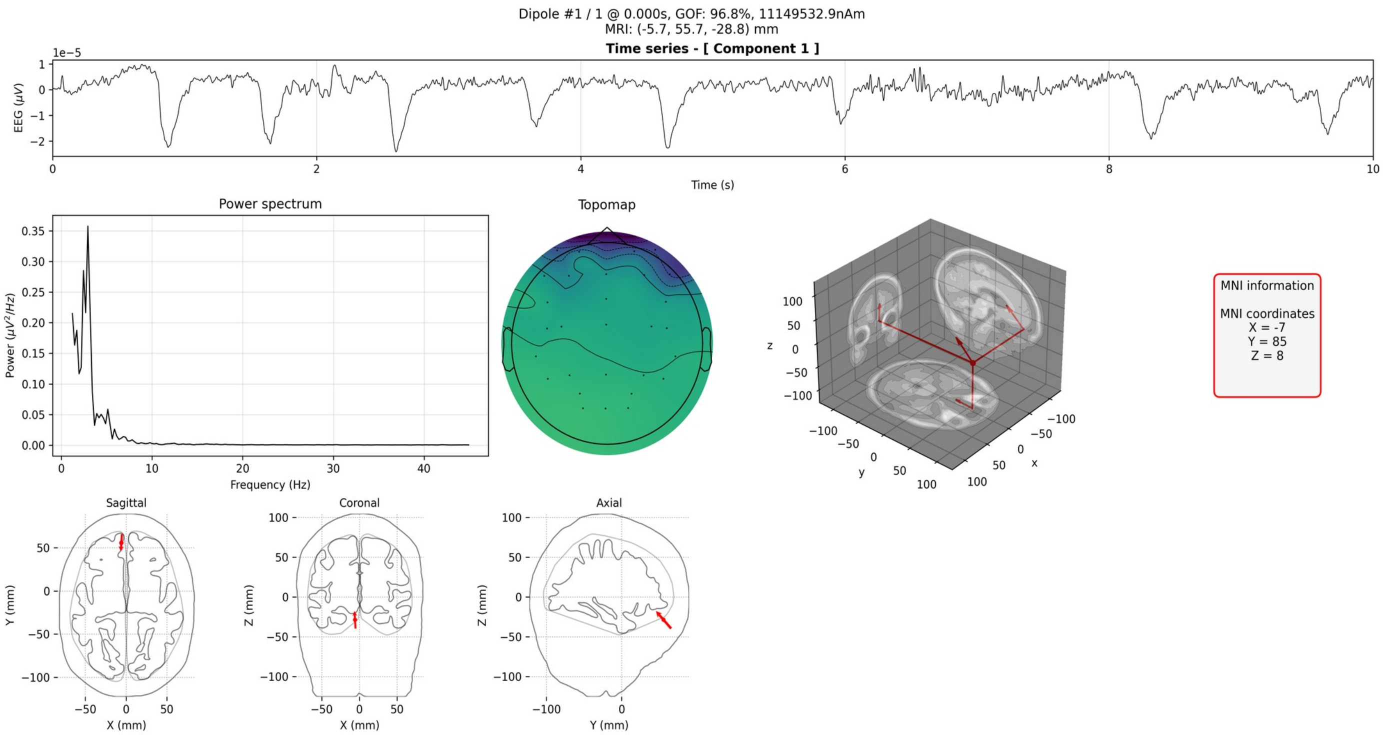


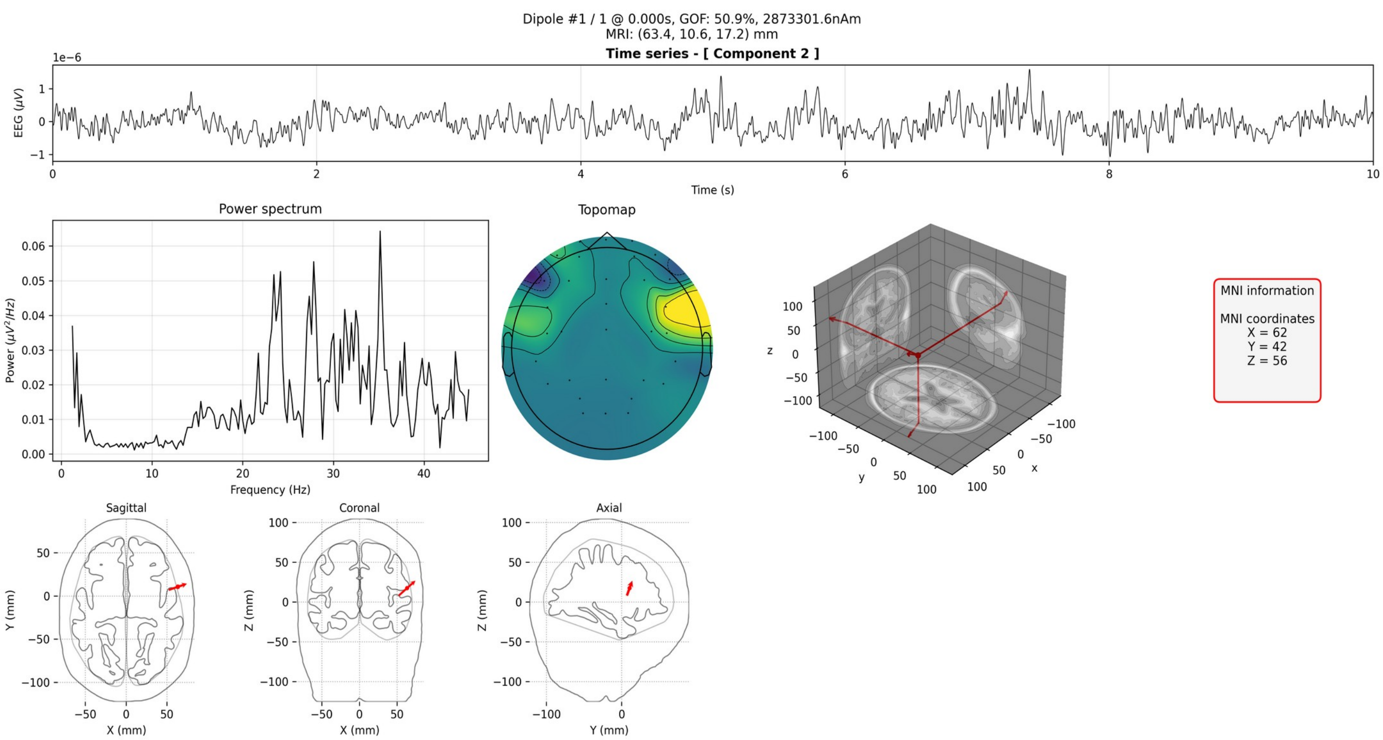


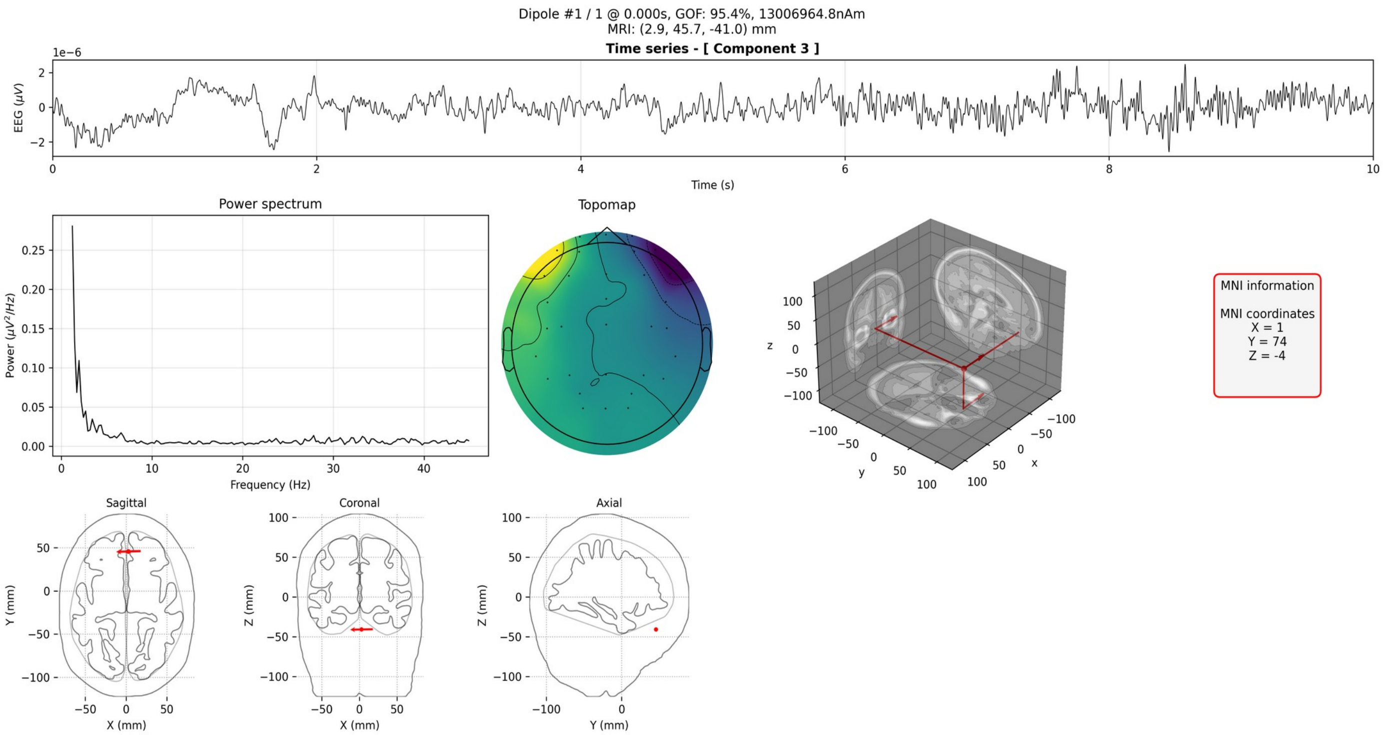


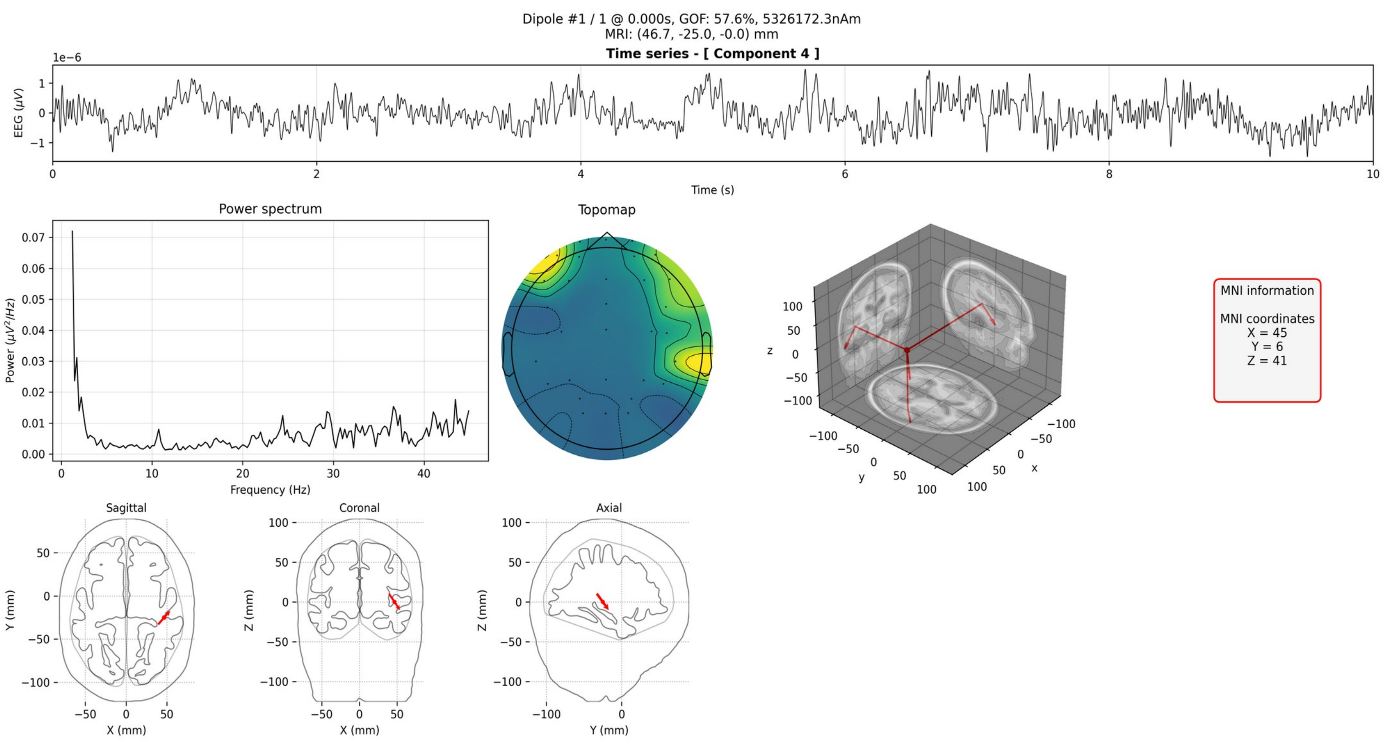
III. ICA Component Analysis

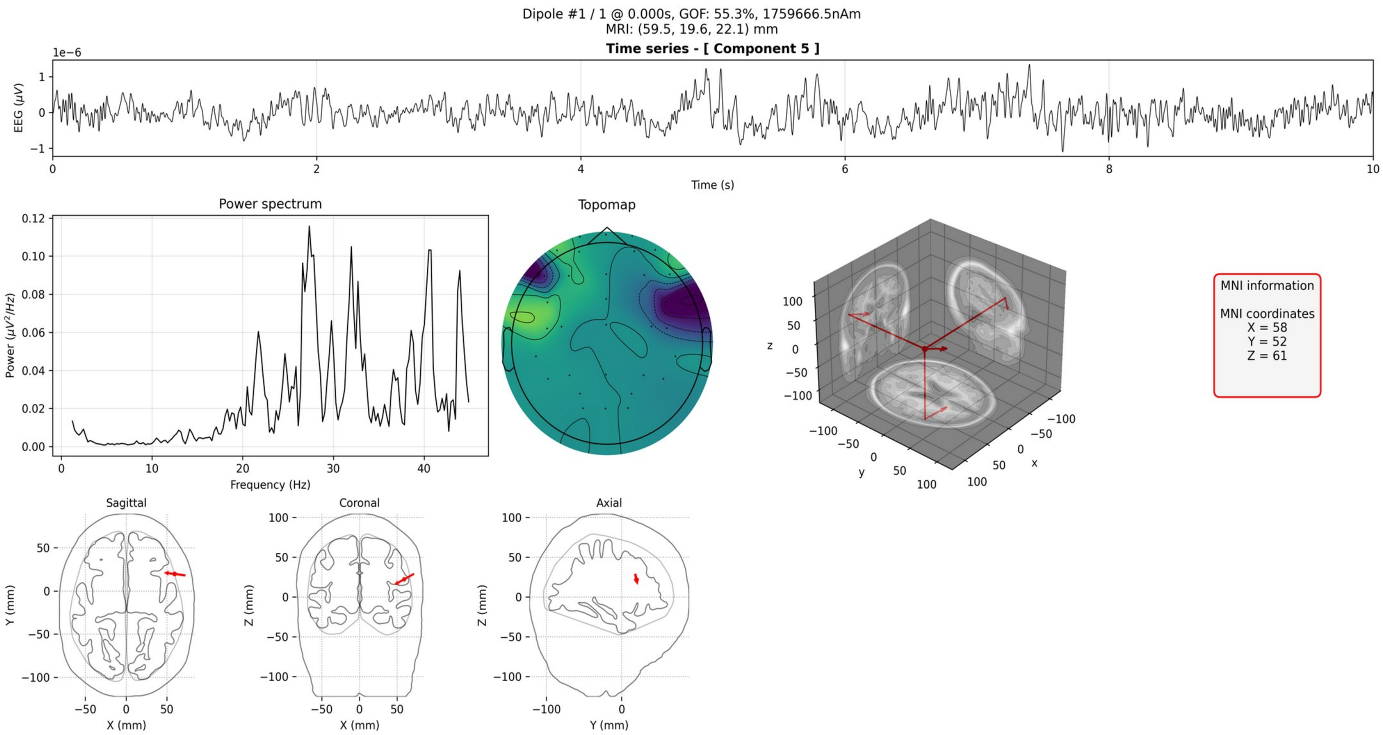
Independent component analysis (ICA) is a statistical method to separate independent sources from superimposed signals. It is the most common method that has been used in EEG data decomposition, and can be used to identify and remove the artifacts from raw EEG data. Features including time series, power spectrum density (PSD), component scalp map (Topomap), dipole source location (Source location) extracted from ICA are shown for each component.

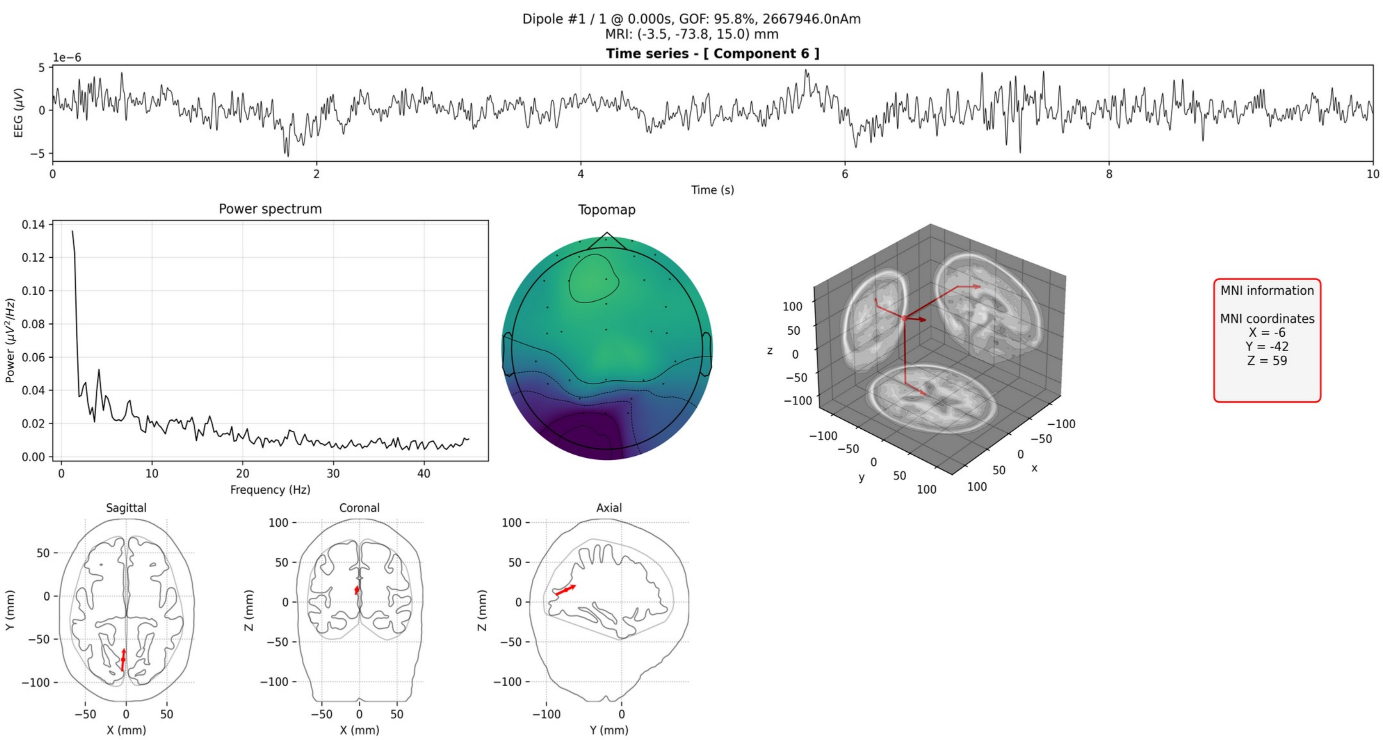

Dipole #1 / 1 @ 0.000s, GOF: 96.8%, 11149532.9nAm
MRI: (-5.7, 55.7, -28.8) mm
Time series - [Component 1]
1e−5
EEG (μV)
Time (s)
Power spectrum
Power (μV²/Hz)
Frequency (Hz)
Topomap
MNI information
MNI coordinates
X = -7
Y = 85
Z = 8
Sagittal
Y (mm)
X (mm)
Coronal
Z (mm)
X (mm)
Axial
Z (mm)
Y (mm)

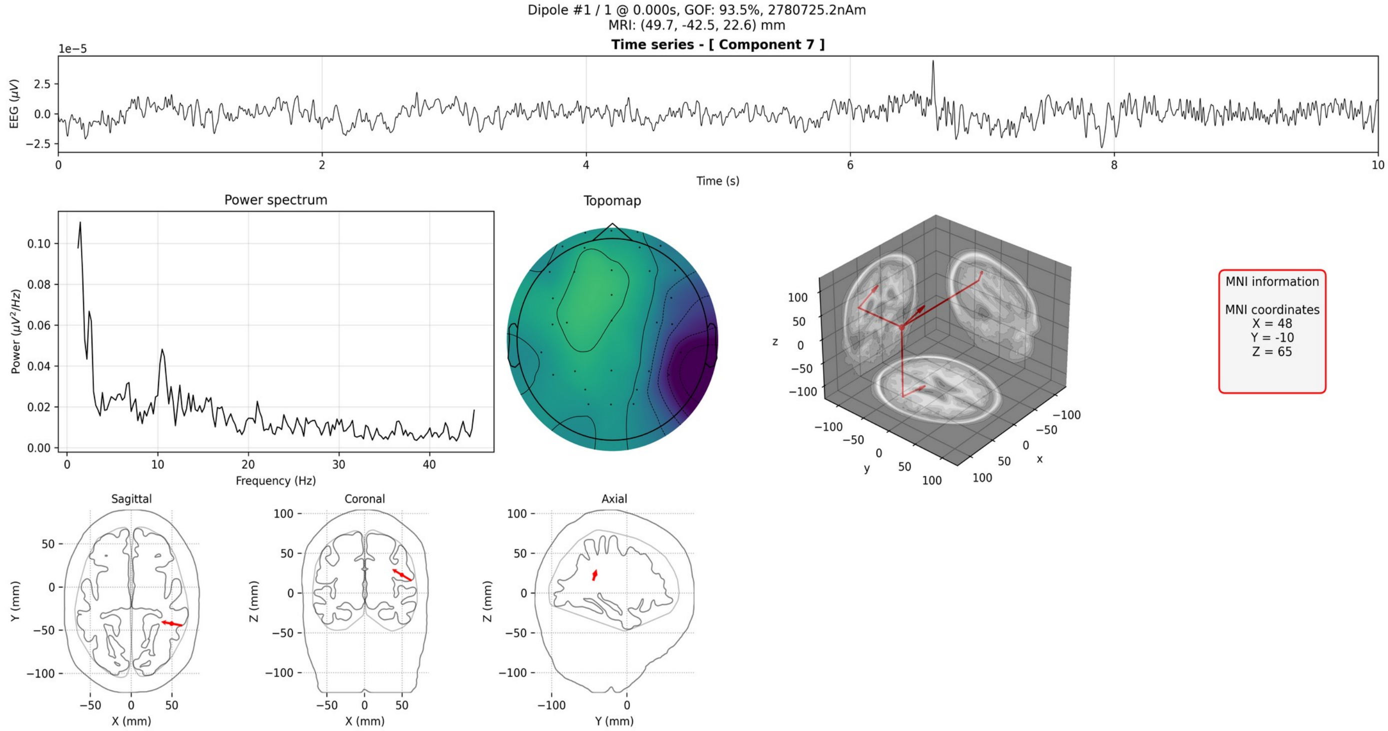

Dipole #1 / 1 @ 0.000s, GOF: 50.9%, 2873301.6nAm
MRI: (63.4, 10.6, 17.2) mm
Time series - [Component 2]
1e−6
EEG (μV)
Time (s)
Power spectrum
Power (μV²/Hz)
Frequency (Hz)
Topomap
MNI information
MNI coordinates
X = 62
Y = 42
Z = 56
Sagittal
Y (mm)
X (mm)
Coronal
Z (mm)
X (mm)
Axial
Z (mm)
Y (mm)

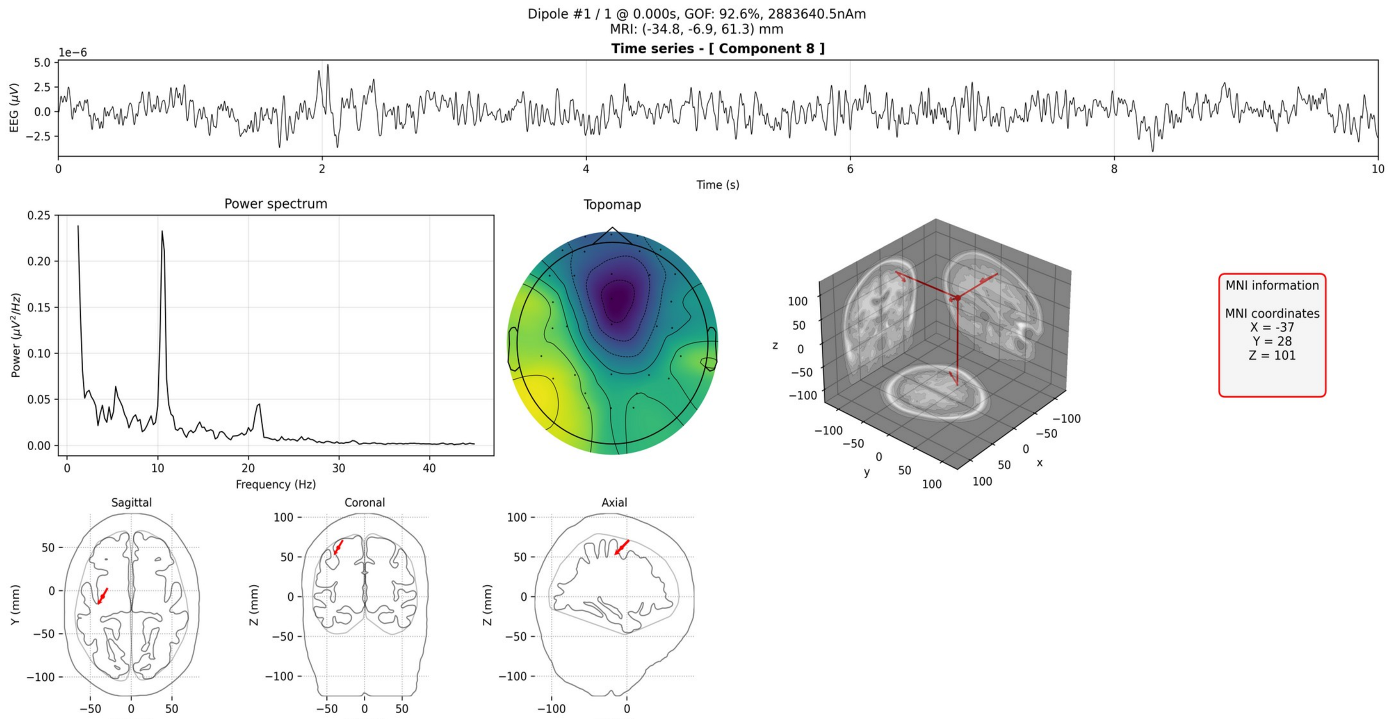

Dipole #1 / 1 @ 0.000s, GOF: 95.4%, 13006964.8nAm
MRI: (2.9, 45.7, -41.0) mm
Time series - [Component 3]
EEG (μV)
Time (s)
Power spectrum
Power (μV²/Hz)
Frequency (Hz)
Topomap
MNI information
MNI coordinates
X = 1
Y = 74
Z = -4
Sagittal
Coronal
Axial
Y (mm)
X (mm)
Z (mm)
Y (mm)

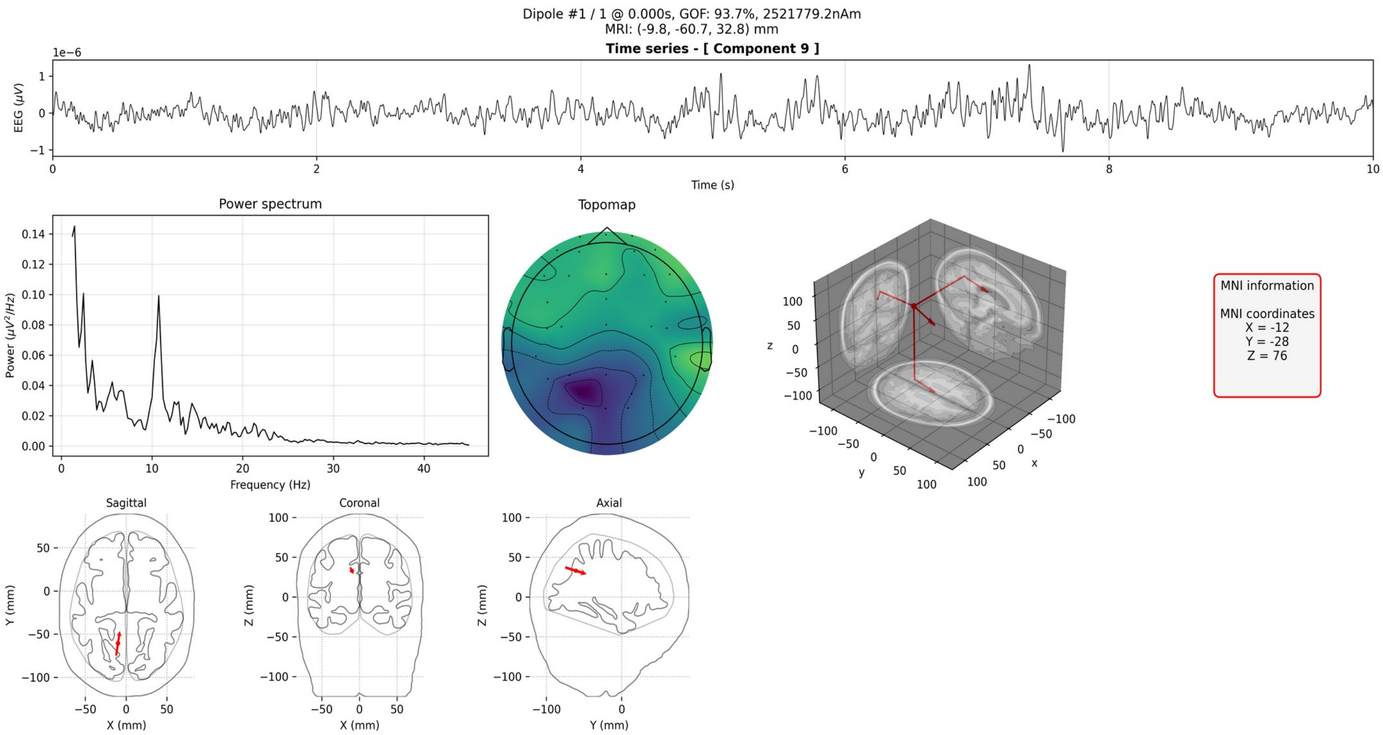

Dipole #1 / 1 @ 0.000s, GOF: 57.6%, 5326172.3nAm
MRI: (46.7, -25.0, -0.0) mm
Time series - [Component 4]
EEG (μV)
Time (s)
Power spectrum
Power (μV²/Hz)
Frequency (Hz)
Topomap
MNI information
MNI coordinates
X = 45
Y = 6
Z = 41
Sagittal
Coronal
Axial
Y (mm)
X (mm)
Z (mm)
Y (mm)

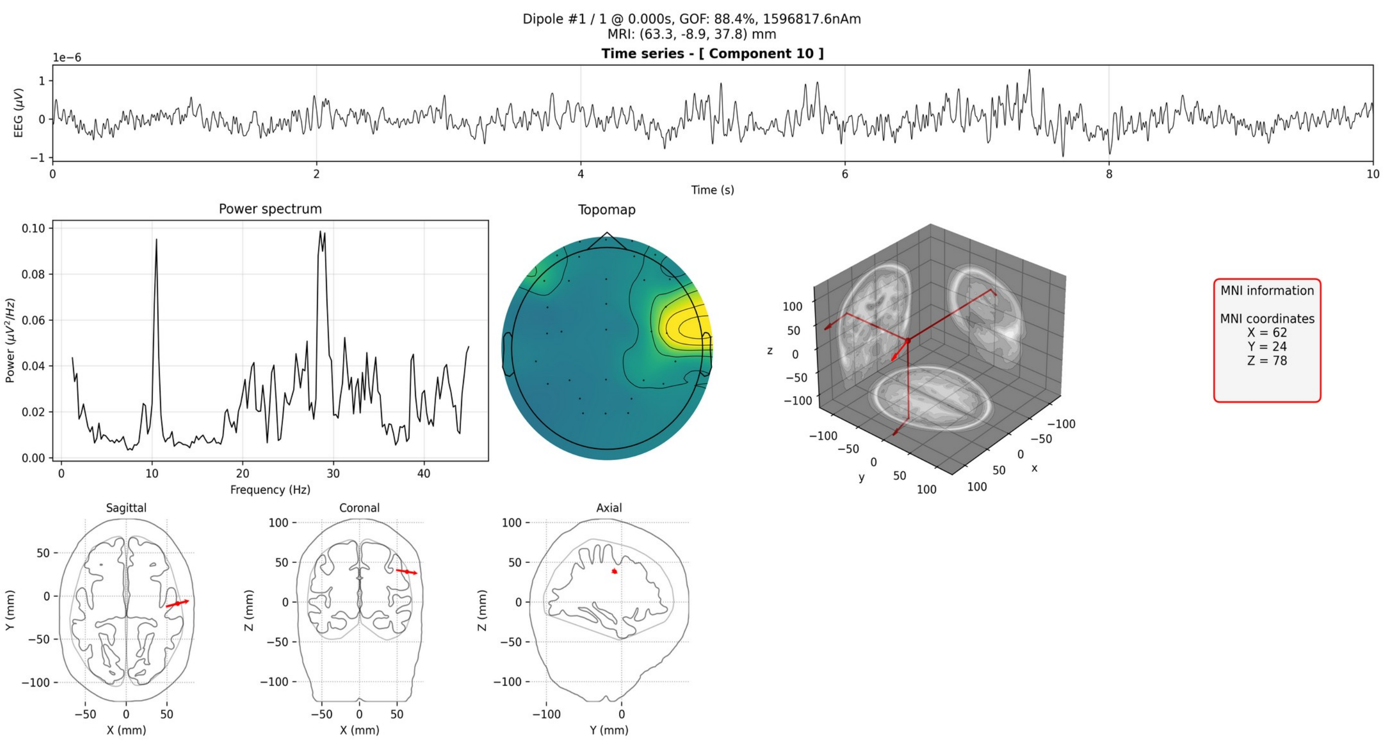

Dipole #1 / 1 @ 0.000s, GOF: 55.3%, 1759666.5nAm
MRI: (59.5, 19.6, 22.1) mm
Time series - [Component 5]
EEG (μV)
Time (s)
Power spectrum
Power (μV²/Hz)
Frequency (Hz)
Topomap
MNI information
MNI coordinates
X = 58
Y = 52
Z = 61
Sagittal
Coronal
Axial
Y (mm)
X (mm)
Z (mm)

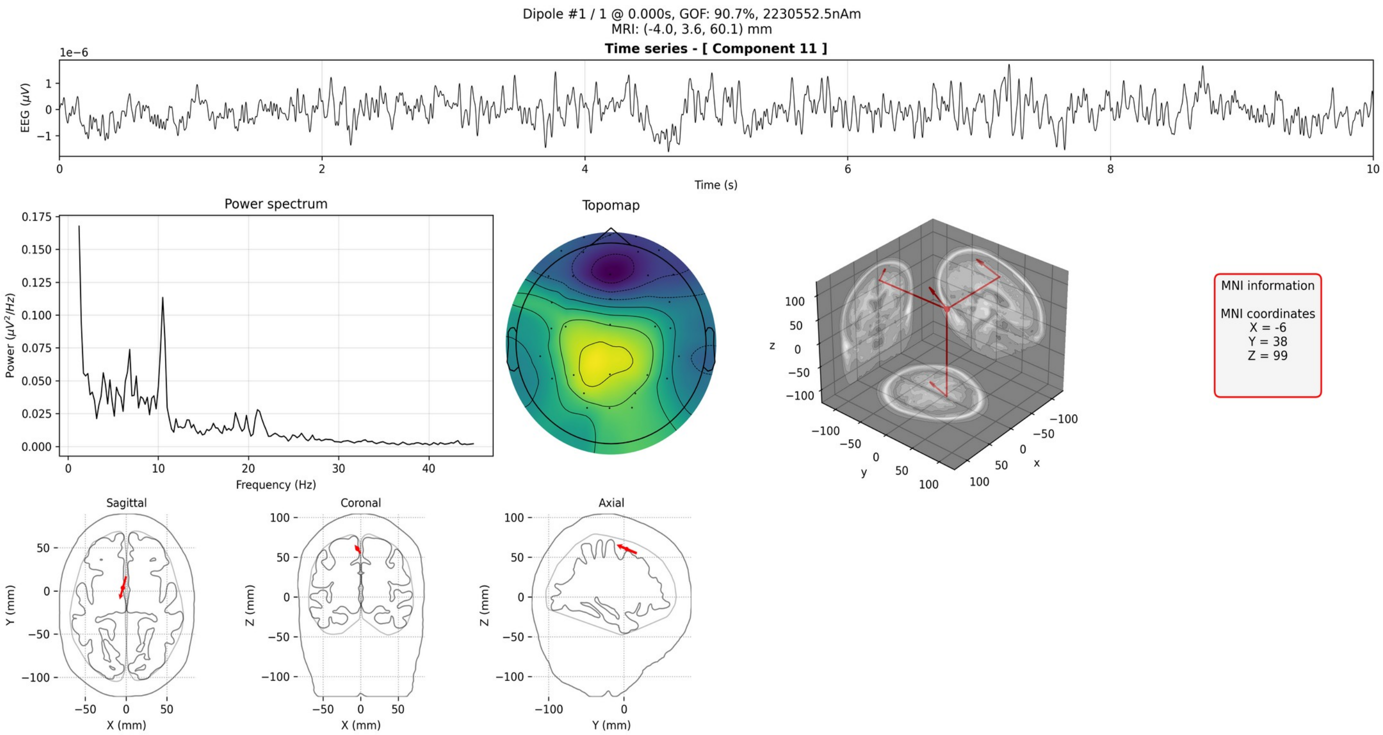

Dipole #1 / 1 @ 0.000s, GOF: 95.8%, 2667946.0nAm
MRI: (-3.5, -73.8, 15.0) mm
Time series - [Component 6]
EEG (μV)
Time (s)
Power spectrum
Power (μV²/Hz)
Frequency (Hz)
Topomap
MNI information
MNI coordinates
X = -6
Y = -42
Z = 59
Sagittal
Coronal
Axial
Y (mm)
X (mm)
Z (mm)

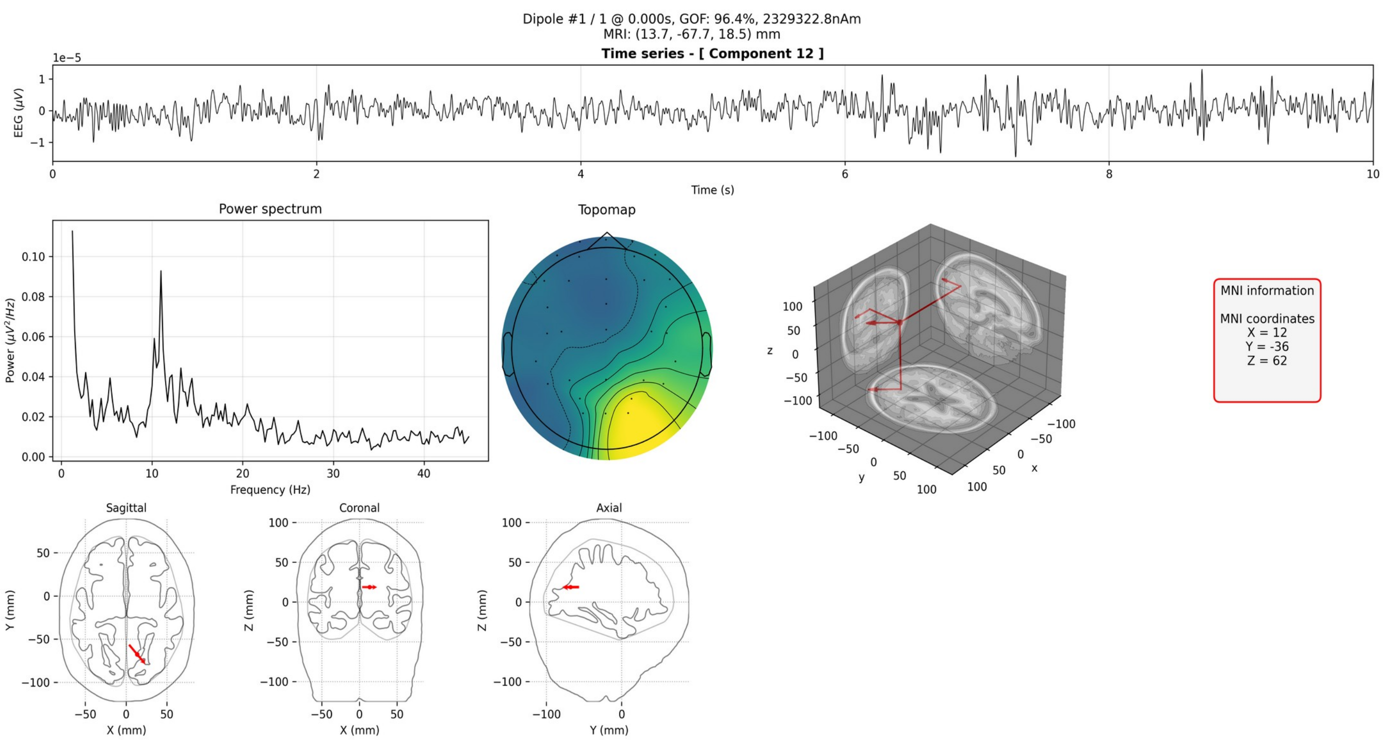

Dipole #1 / 1 @ 0.000s, GOF: 93.5%, 2780725.2nAm
MRI: (49.7, -42.5, 22.6) mm
Time series - [Component 7]
Power spectrum
Topomap
MNI information
MNI coordinates
X = 48
Y = -10
Z = 65
Sagittal
Coronal
Axial

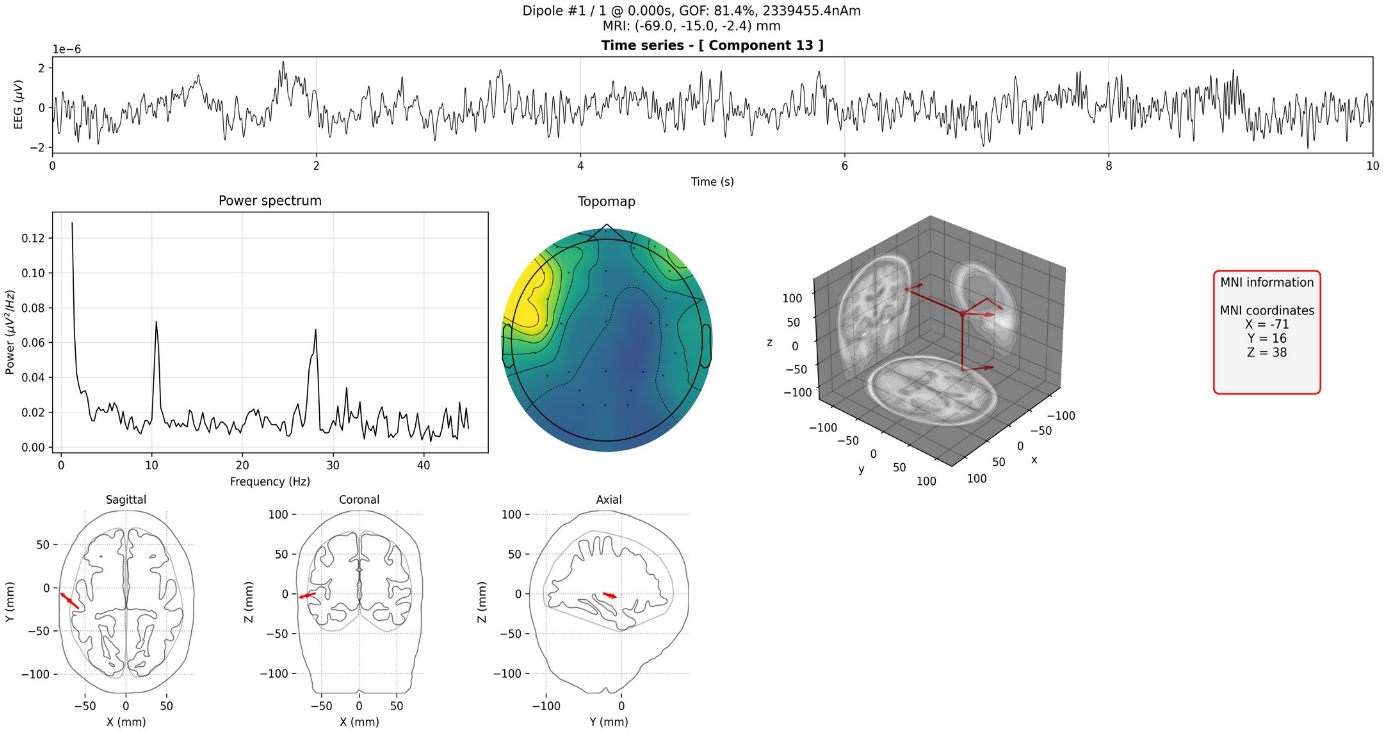

Dipole #1 / 1 @ 0.000s, GOF: 92.6%, 2883640.5nAm
MRI: (-34.8, -6.9, 61.3) mm
Time series - [Component 8]
Power spectrum
Topomap
MNI information
MNI coordinates
X = -37
Y = 28
Z = 101
Sagittal
Coronal
Axial

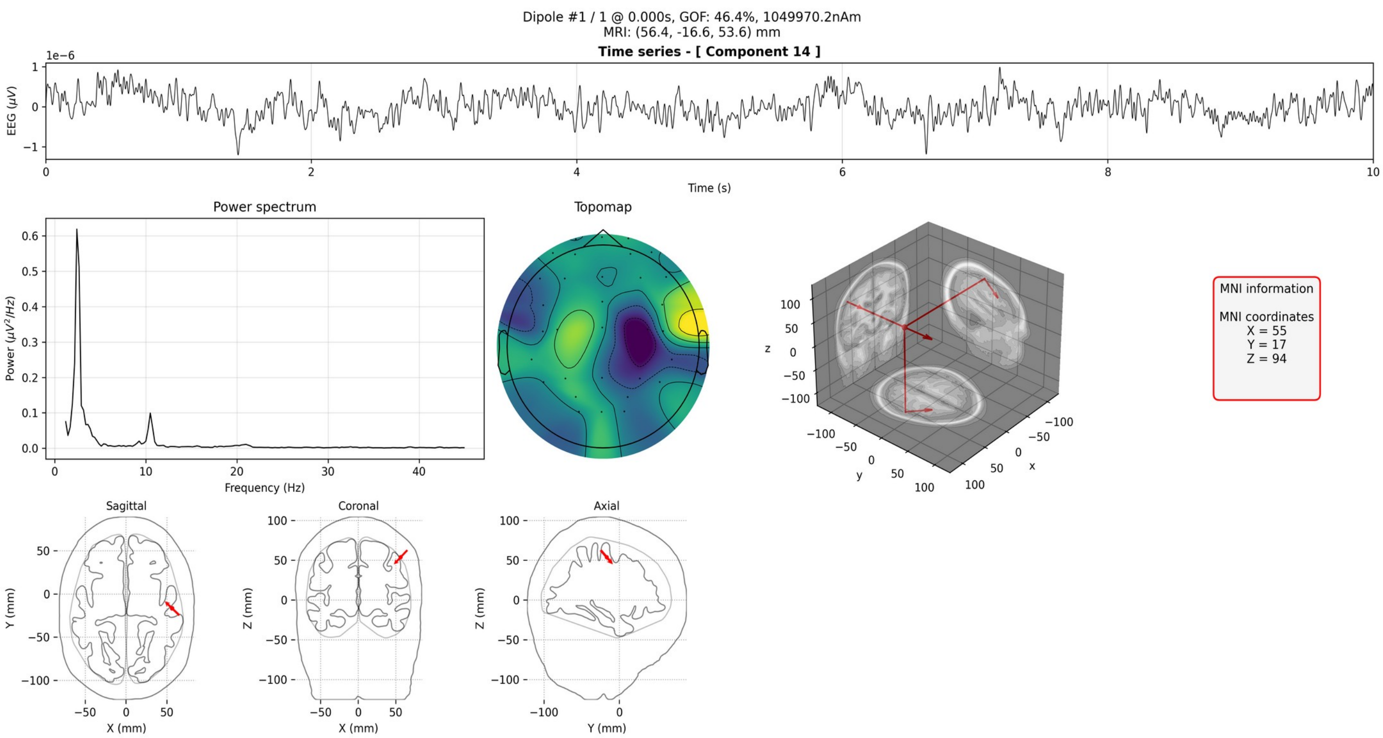

Dipole #1 / 1 @ 0.000s, GOF: 93.7%, 2521779.2nAm
MRI: (-9.8, -60.7, 32.8) mm
Time series - [Component 9]
EEG (μV)
Time (s)
Power spectrum
Power (μV²/Hz)
Frequency (Hz)
Topomap
MNI information
MNI coordinates
X = -12
Y = -28
Z = 76
Sagittal
Coronal
Axial
Y (mm)
X (mm)
Z (mm)

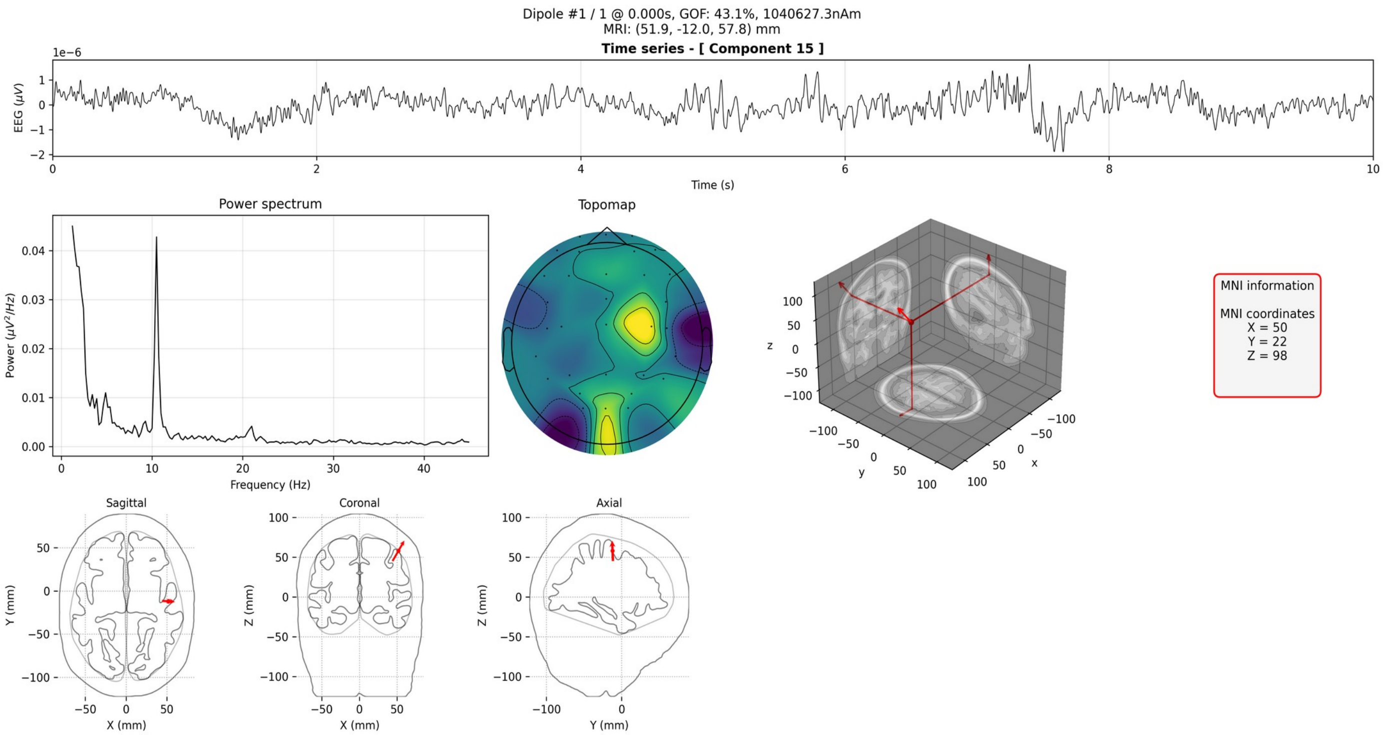

Dipole #1 / 1 @ 0.000s, GOF: 88.4%, 1596817.6nAm
MRI: (63.3, -8.9, 37.8) mm
Time series - [Component 10]
EEG (μV)
Time (s)
Power spectrum
Power (μV²/Hz)
Frequency (Hz)
Topomap
MNI information
MNI coordinates
X = 62
Y = 24
Z = 78
Sagittal
Coronal
Axial
Y (mm)
X (mm)
Z (mm)

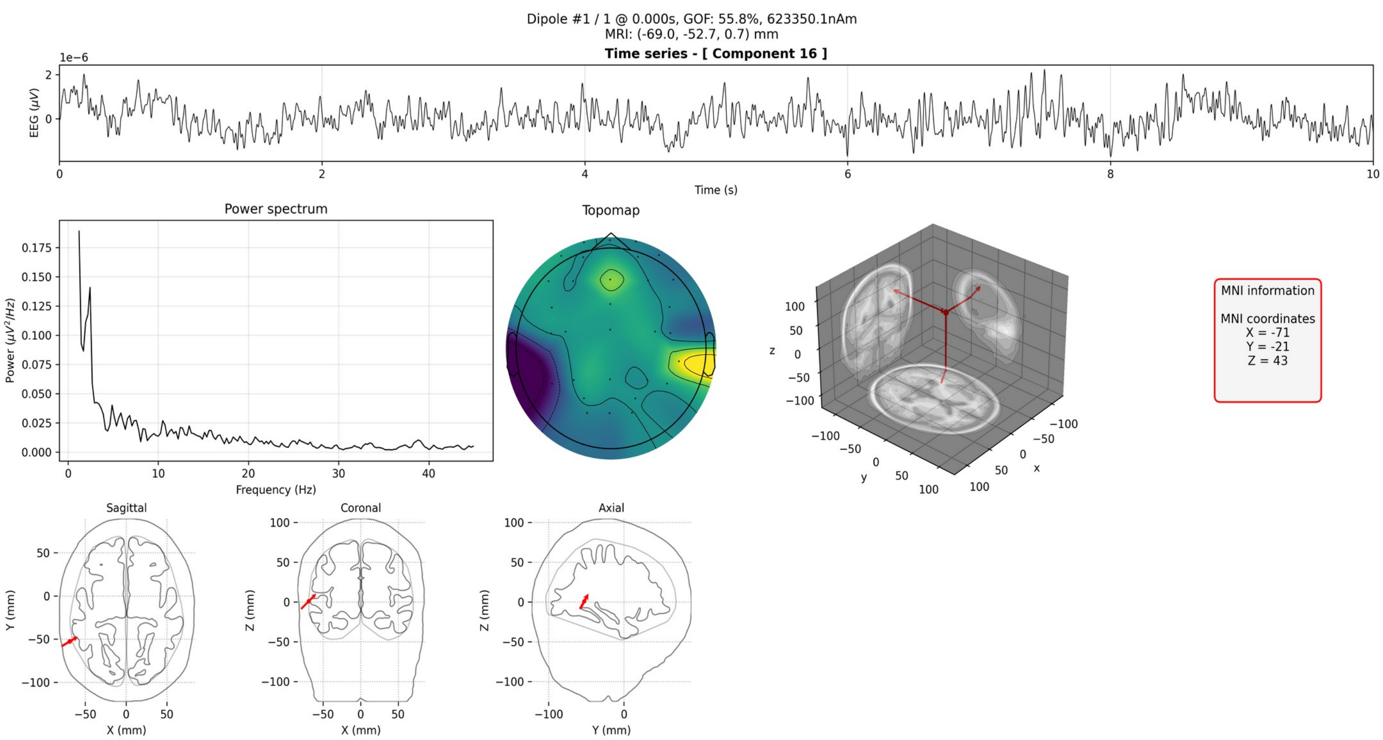

Dipole #1 / 1 @ 0.000s, GOF: 90.7%, 2230552.5nAm
MRI: (-4.0, 3.6, 60.1) mm
Time series - [Component 11]
EEG (μV)
Time (s)
Power spectrum
Power (μV²/Hz)
Frequency (Hz)
Topomap
MNI information
MNI coordinates
X = -6
Y = 38
Z = 99
Sagittal
Coronal
Axial
Y (mm)
Z (mm)
X (mm)

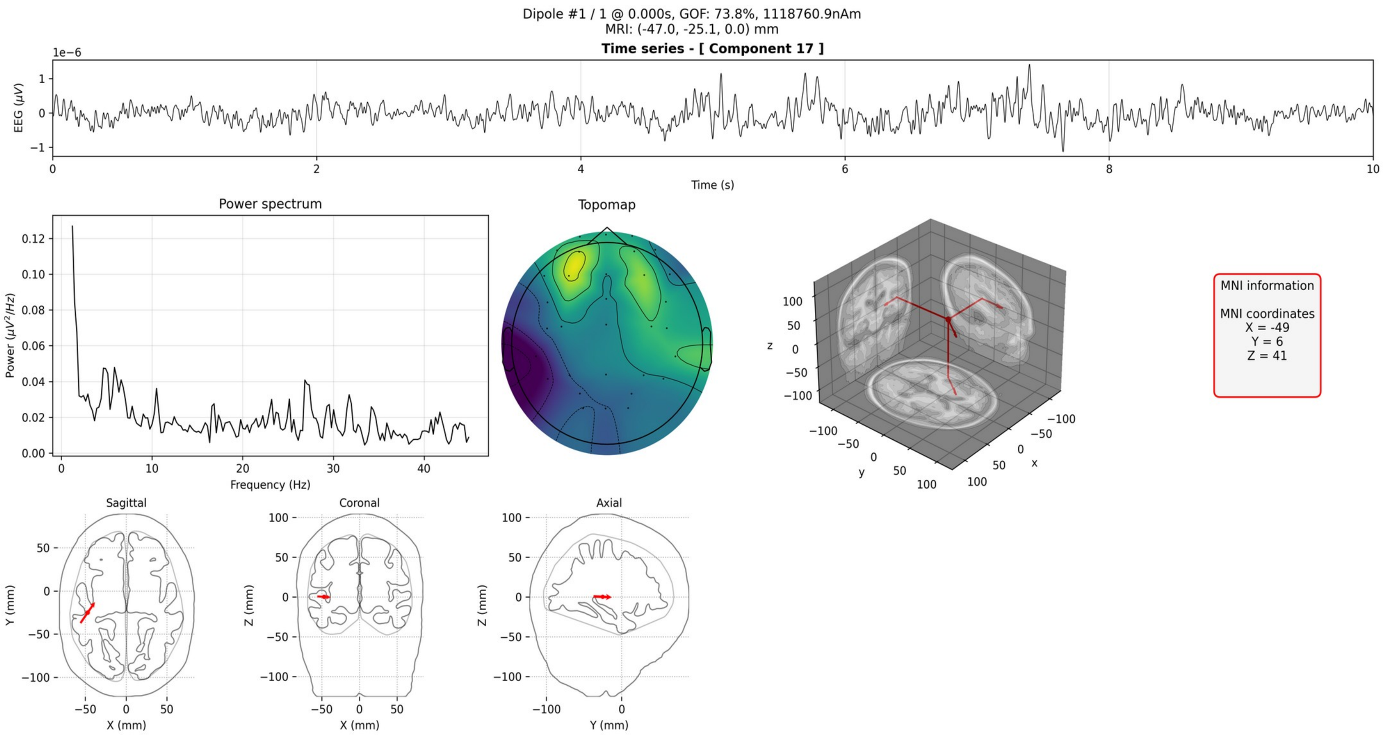

Dipole #1 / 1 @ 0.000s, GOF: 96.4%, 2329322.8nAm
MRI: (13.7, -67.7, 18.5) mm
Time series - [Component 12]
EEG (μV)
Time (s)
Power spectrum
Power (μV²/Hz)
Frequency (Hz)
Topomap
MNI information
MNI coordinates
X = 12
Y = -36
Z = 62
Sagittal
Coronal
Axial
Y (mm)
Z (mm)
X (mm)

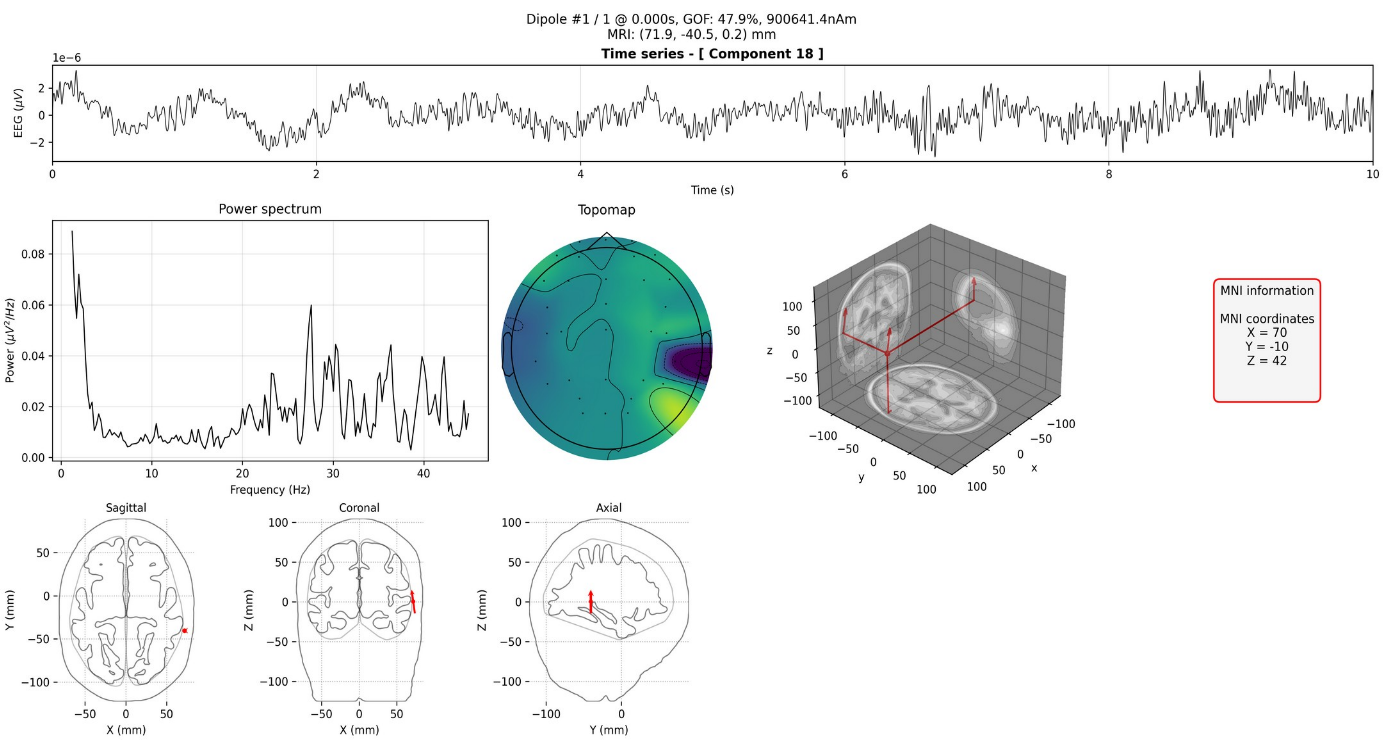

Dipole #1 / 1 @ 0.000s, GOF: 81.4%, 2339455.4nAm
MRI: (-69.0, -15.0, -2.4) mm
Time series - [Component 13]
EEG (µV)
Time (s)
Power spectrum
Power (µV²/Hz)
Frequency (Hz)
Topomap
MNI information
MNI coordinates
X = -71
Y = 16
Z = 38
Sagittal
Coronal
Axial
Y (mm)
X (mm)
Z (mm)

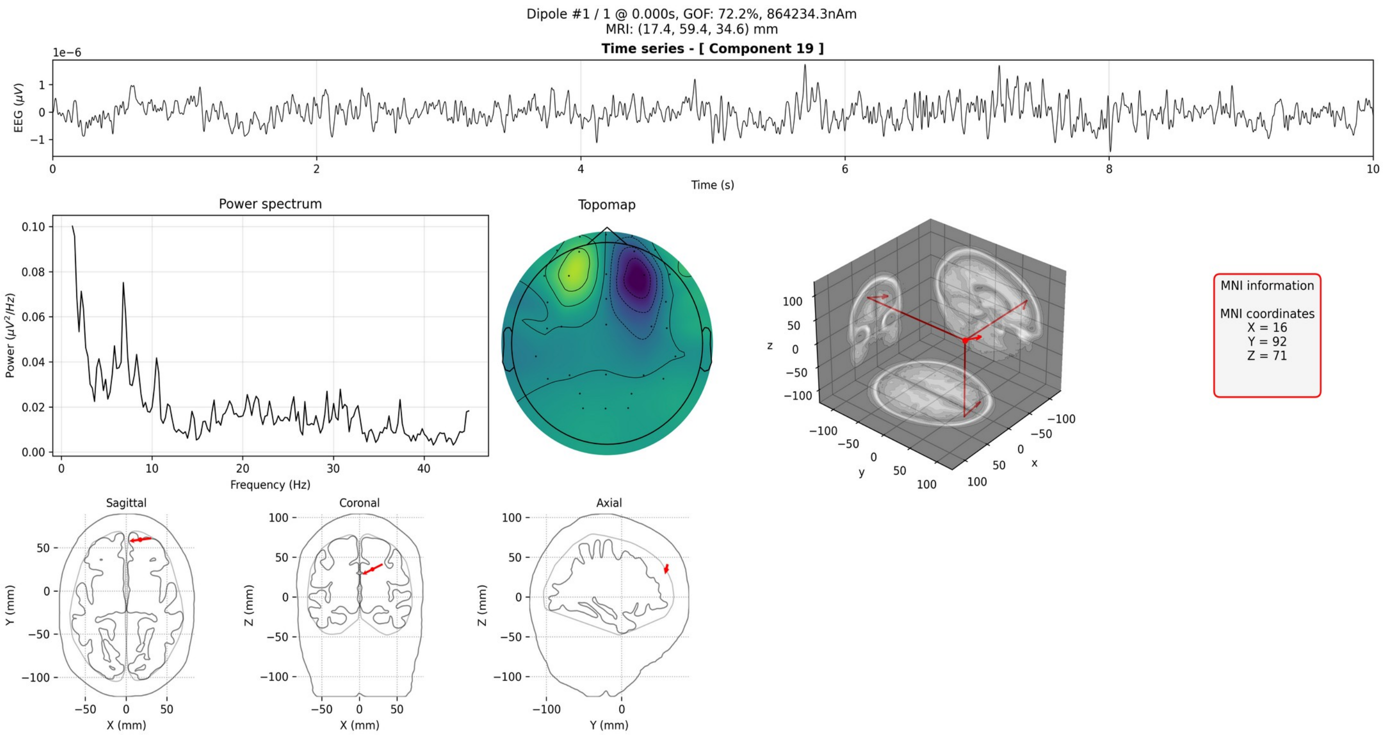

Dipole #1 / 1 @ 0.000s, GOF: 46.4%, 1049970.2nAm
MRI: (56.4, -16.6, 53.6) mm
Time series - [Component 14]
EEG (µV)
Time (s)
Power spectrum
Power (µV²/Hz)
Frequency (Hz)
Topomap
MNI information
MNI coordinates
X = 55
Y = 17
Z = 94
Sagittal
Coronal
Axial
Y (mm)
X (mm)
Z (mm)


Dipole #1 / 1 @ 0.000s, GOF: 43.1%, 1040627.3nAm
MRI: (51.9, -12.0, 57.8) mm
Time series - [Component 15]
EEG (µV)
Time (s)
Power spectrum
Power (µV²/Hz)
Frequency (Hz)
Topomap
MNI information
MNI coordinates
X = 50
Y = 22
Z = 98
Sagittal
Coronal
Axial
X (mm)
Y (mm)
Z (mm)


Dipole #1 / 1 @ 0.000s, GOF: 55.8%, 623350.1nAm
MRI: (-69.0, -52.7, 0.7) mm
Time series - [Component 16]
EEG (µV)
Time (s)
Power spectrum
Power (µV²/Hz)
Frequency (Hz)
Topomap
MNI information
MNI coordinates
X = -71
Y = -21
Z = 43
Sagittal
Coronal
Axial
X (mm)
Y (mm)
Z (mm)


Dipole #1 / 1 @ 0.000s, GOF: 73.8%, 1118760.9nAm
MRI: (-47.0, -25.1, 0.0) mm
Time series - [Component 17]
EEG (μV)
Time (s)
Power spectrum
Power (μV²/Hz)
Frequency (Hz)
Topomap
MNI information
MNI coordinates
X = -49
Y = 6
Z = 41
Sagittal
Coronal
Axial
X (mm)
Y (mm)
Z (mm)

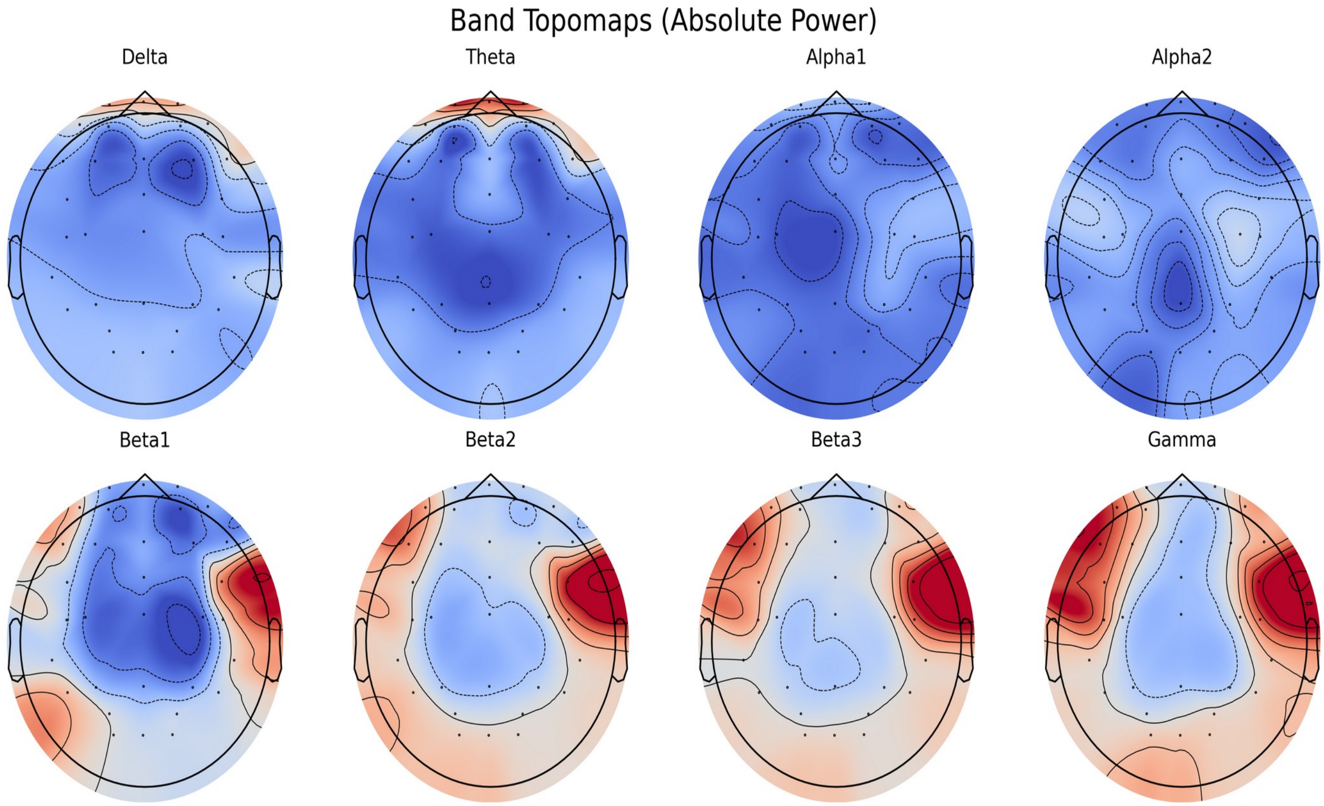

Dipole #1 / 1 @ 0.000s, GOF: 47.9%, 900641.4nAm
MRI: (71.9, -40.5, 0.2) mm
Time series - [Component 18]
EEG (μV)
Time (s)
Power spectrum
Power (μV²/Hz)
Frequency (Hz)
Topomap
MNI information
MNI coordinates
X = 70
Y = -10
Z = 42
Sagittal
Coronal
Axial
X (mm)
Y (mm)
Z (mm)

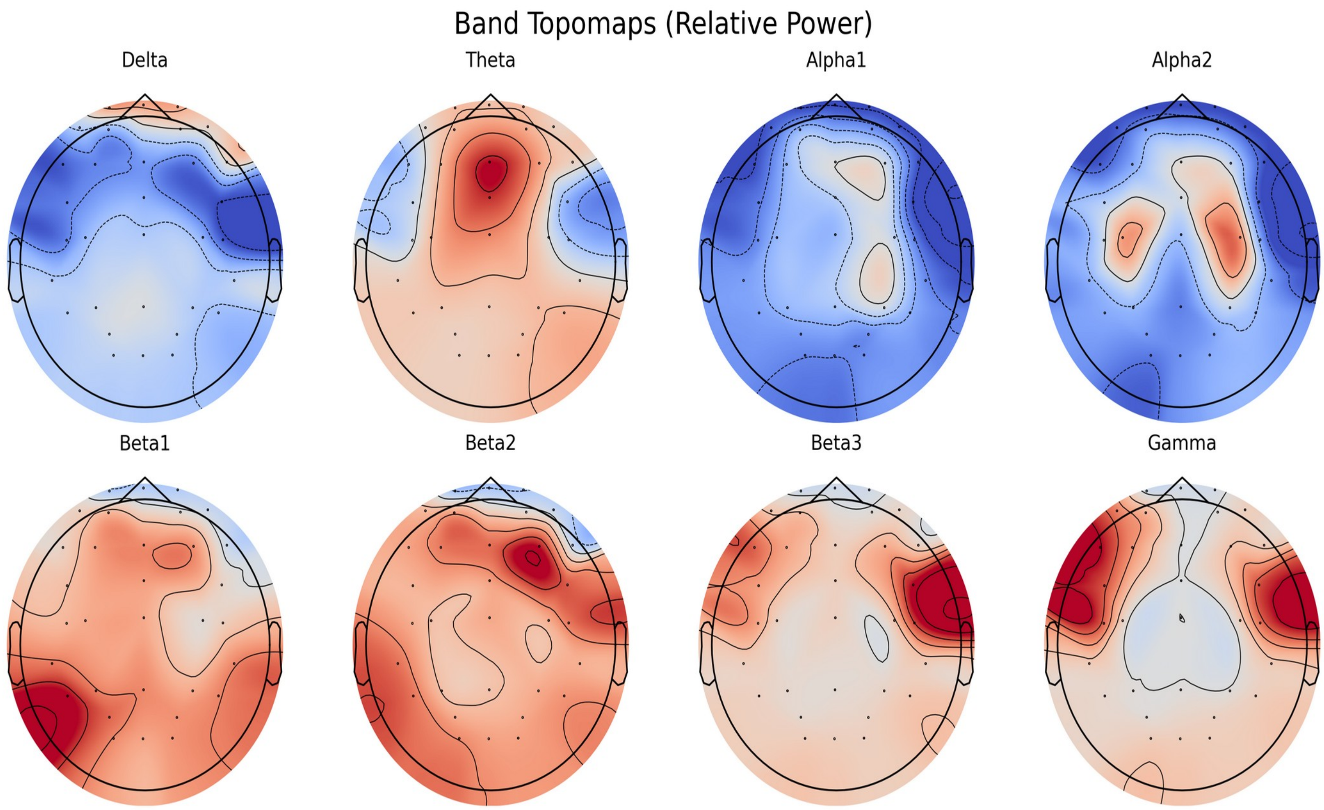


IV. Band power - Topomap

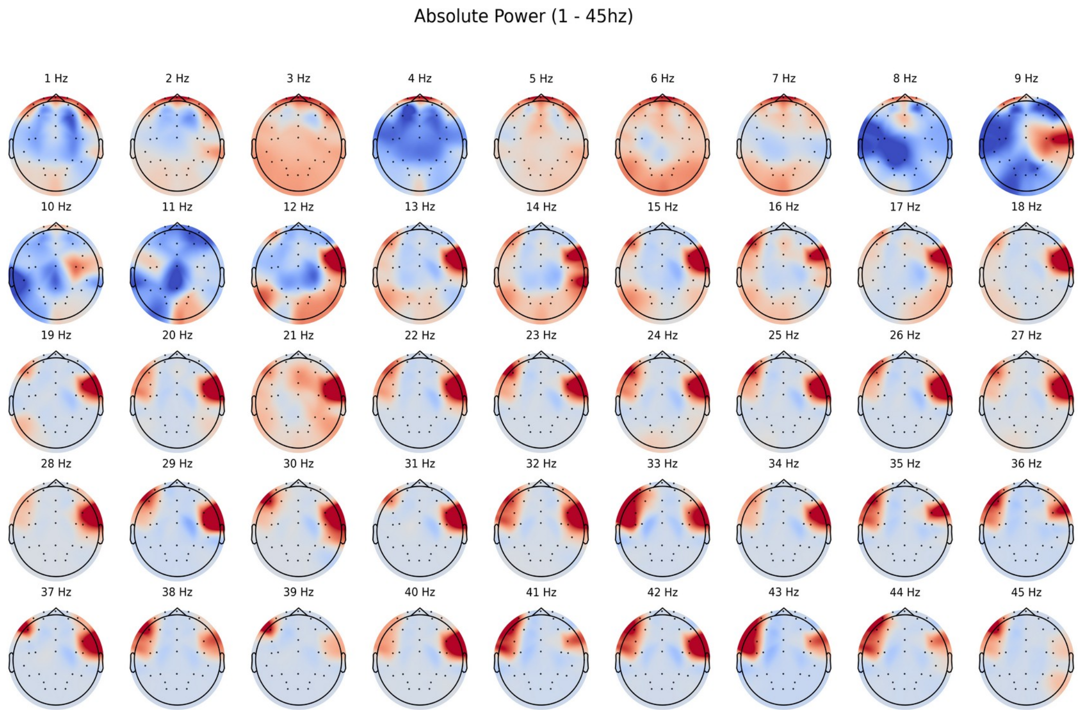
In power spectral density (PSD) 2D map, topomaps of absolute and relative power in 1 Hz bins (1 - 45 Hz) as well as each frequency band are presented. Absolute power is the sum of the component powers for each frequency band. Relative power is the absolute power in a specific frequency band divided by the total power. It is advisable to compare relative power with absolute power, since absolute power reflects the individual differences due to variations in brain tissue. This feature provides absolute and relative power based on six brain regions (prefrontal, frontal, left temporal, right temporal, central, parietal, and occipital). The power spectra for each of the 19 channels are shown in the following feature, PSD spectrum (below).

IV-I Band


Band Topomaps (Absolute Power)
Delta
Theta
Alpha1
Alpha2
Beta1
Beta2
Beta3
Gamma


Band Topomaps (Relative Power)
Delta
Theta
Alpha1
Alpha2
Beta1
Beta2
Beta3
Gamma

IV-II Absolute band powers (1 – 45 Hz)



IV-III Relative band powers (1 – 45 Hz)